MTH 200 Calculating Limits graphically

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider $\lim_{x \to 0} \frac{e^x - 1}{x}$.

a) Set your calculator to a standard 10x10 graphing window. What does the graph of this function suggest the limit would be?

b) Zoom in on the graph near $x = 0$. What does the graph now suggest is the value of the limit?

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c) What does the table of function values suggest for the value of the limit?

d) Do the previous parts confirm that the given limit equals 1?

2. Consider $\lim\limits_{x\to 0}\dfrac{46\left[1-\cos(3x)\right]}{45x^2}$.

a) Set your calculator to a standard 10x10 graphing window. What does the graph of this function suggest the limit would be?

b) Zoom in on the graph near $x=0$. What does the graph now suggest is the value of the limit?

c) Use the fact that $\lim\limits_{\theta\to 0}\dfrac{\sin(\theta)}{\theta}=1$ to evaluate the limit in this question. *Hint:* multiply the numerator and denominator by $\left[1+\cos(3x)\right]$.

3. Consider $\lim_{x\to-\infty} x\left(\sqrt{x^2+1}+x\right)$. Since we are consider the limit as x decreases without bound, we will want to zoom out and consider larger and larger negative values on the x-axis.

a) As you initially begin to zoom out on the graph (values between -2×10^6 and 0), what do you believe the value of this limit is? Attempt to adjust your window to have ymin = -1 and ymax = 1.

b) What is initially suggest by the table of values?

c) What does the graph suggest about the limit when you consider values in the range -5×10^7 to -2×10^6? What does the table of values suggest?

d) Algebraically determine the correct value of the limit. *Hint:* consider multiplying by a "fancy 1".

4. Based on the above problems, explain why we should be cautious when using technology to evaluate limits.

5. Based on your above observations, explain why we need to develop mathematical techniques to evaluate limits.